

Context-Aware Speed and Separation Monitoring for Human–Robot Ceiling-Panel Installation: An Exploratory Evaluation

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Abstract—In ceiling-panel installation, a worker and a robot share a workspace while handling the same panel. Fixed safety zones cannot tell whether the worker is approaching, working or moving away, so they can stop the robot more often than needed. This paper presents a speed and separation monitoring (SSM) controller that uses the task phase, and tests it on a physical prototype. A hidden Markov model (HMM) estimates the task phase from motion-capture data, and a separate predictor detects fast movement toward the robot. A fixed rule ensures that neither of these learned parts can make the robot faster than the basic distance-based limit allows. Tested on operators it had not been trained on, the real-time phase model was correct 79.4% of the time (balanced accuracy 72.5%). The study produced 97 usable robot-data trials from 11 study codes under fixed, reactive and predictive control, plus 31 block questionnaires. Participants were confident that the robot would stop and reported low workload, and their average safety rating rose from 3.82/5 before the session to 4.45/5 after it. We hypothesised that the predictive controller is as safe as fixed-zone control, the current industry approach, while keeping comparable efficiency. The safety indicators matched: every controller requested a stop whenever a person was inside the red boundary, and closest approach and time inside it were similar. Trials took about as long, but the predictive controller kept the robot stopped for more of each trial, so the hypothesis is only partly supported. The study shows that task-phase recognition, intrusion prediction and two motion-capture systems can work together in a construction-style task, and it sets out what must be measured to compare the three controllers conclusively.

Index Terms—Human–robot collaboration, speed and separation monitoring, hidden Markov model, construction robotics, motion capture, human factors.

I. INTRODUCTION AND BACKGROUND

INSTALLING ceiling panels involves lifting, aligning and fixing a panel overhead (Fig. 1). A robot can hold the panel while the worker provides judgement and fine hand skills. However, the worker moves around during the task, the panel enlarges the space the robot occupies, and unnecessary robot stops can break the flow of work.

Speed and separation monitoring (SSM) keeps enough distance between a person and a robot, and slows or stops the robot in time [1]–[3]. The simplest form uses fixed distance thresholds. These are easy to understand, but they treat a person walking toward the robot, standing still and walking away in the same way. Knowing the task phase could help the

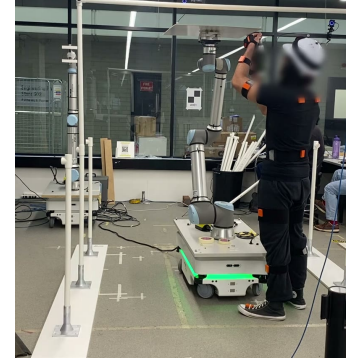


Fig. 1. The ceiling-panel installation task in our lab: the robot holds the panel at ceiling height while the worker fixes it from below (frame from session video; face blurred).

robot avoid unnecessary stops, as long as a wrongly recognised phase can never loosen the safety limit. Our controller therefore keeps three parts separate: task-phase recognition, motion prediction and the distance limit.

The research question is whether a controller that knows the task phase can keep people as safe as the fixed-zone control used in practice today, while keeping the task efficient. The planned comparison covers response to intrusion, unnecessary stops, task efficiency and participant experience. This exploratory analysis answers three supporting questions: (1) how accurately the model recognises the task phase in real time; (2) how the distance measurement and the scripted movements affect how the recorded robot behaviour should be read; and (3) what the questionnaires show about participants' experience across the task blocks. Our main hypothesis (H1) is that the context-aware predictive controller is as safe as fixed-zone control, the current industry approach, while maintaining comparable task efficiency. Its safety rests on the same distance limit: the learned parts are not allowed to relax it.

This work makes three contributions: (1) a physical ceiling-panel prototype in which task-phase recognition and intrusion prediction are kept separate, and both sit below a fixed speed limit; (2) a descriptive analysis of the distance inputs and robot motion recorded during the task; and (3) item-by-item questionnaire results, checked again using only complete response sets. The contribution is combining and testing these parts in a construction task, not a new SSM formula. The rest of this paper is organised as follows. Section II reviews related work, and Section III describes the prototype and its controllers.

Section IV reports how the phase model was developed and validated, and Section V describes the experiment. Section VI presents the results, Section VII discusses them and their limitations, and Section VIII concludes.

II. RELATED WORK

A. Human–Robot Collaboration in Construction

Construction robotics is moving from fully automated machines toward systems that combine human skill with robot strength [4], [5]. Reviews of human–robot collaboration (HRC) in construction list safety, sensing, communication and task sharing as the main barriers to use on site [6], [7]. Panel installation is a useful test case because the person and the robot work around the same object, and stopping too often directly slows the work.

B. Speed and Separation Monitoring

SSM needs more than a fixed safety radius around the robot. The required distance depends on how far the person and the robot move while the system senses, decides and brakes. It also includes margins for body parts that are not tracked and for sensor error [2], [3]. Byner et al. [8] set the robot’s speed limit from the separation distance and direction of motion, and measured cycle performance in a physical machine-tending task. Karagiannis et al. [9] switch between safety-zone layouts depending on where the robot is, using an overhead safety camera, a safety-rated programmable logic controller (PLC) and the robot controller in a cell modelled on the car industry. These studies show that adaptive SSM and its effect on productivity have already been studied. They also define the reactive controller that task context must improve on.

Construction studies narrow the gap further. Xiao Lin et al. [10] combine worker, robot and material shapes in a digital twin that recalculates the protective distance as the scene changes, and test it in static and moving conflict scenarios. Their treatment of carried material matters for panel installation: once the robot carries a panel, a shape drawn around the robot alone no longer covers the space it occupies.

C. Activity Recognition and Collaboration

A hidden Markov model (HMM) tracks a state that changes over time but cannot be observed directly. It uses a transition matrix, which gives how likely the state is to change, and an emission model, which gives the measurements each state tends to produce [11]. HMMs have been used to recognise activities from wearable sensors and to model human–robot tasks [12], [13].

Task phase and hazard are different things. Labels such as “working” or “retreating” describe what the person is doing, while a hazard is about being exposed to harm. A person can trip, or move quickly toward the robot, during any phase. Treating “hazard” as just another phase would mix up activity and danger. We therefore use the HMM only to estimate the task phase. Fast movement toward the robot is detected separately by the intrusion predictor (Section III-D), which

uses only the measured distance and closing speed. Keeping the two apart also means the distance limit can be tested on its own, whatever the HMM decides.

The closest construction study, by Xiao Lin et al. [14], combines a model of the work steps, robot behaviour learned from demonstration and local safety control in a virtual-reality wall-panel task with simulated sensors. Our study adds a physical ceiling-panel prototype with real robot data and participant questionnaires.

How smoothly people and robots work together is usually measured with timing, such as human idle time, robot idle time, time spent working in parallel and delay between actions [15]. Workload and trust questionnaires help explain results, but they cannot prove physical safety or replace measured robot response [16], [17]. This paper therefore links task recognition, measured control inputs and participant experience in one physical panel task. Table I compares these approaches with our prototype.

III. SYSTEM DESIGN

A. Overview

The controller (Fig. 2) has two inputs: the worker’s position and speed relative to the robot, from motion capture, and the position of the robot’s tool, from the robot controller. Its output is a single number between 0 and 1: the fraction of full speed the robot may use. This number is sent to the robot’s speed override, and a value of zero requests a pause.

B. Equipment and Sensing

The lab prototype (Fig. 3) uses a Universal Robots UR10 arm with a vacuum gripper that holds a lightweight stand-in for a ceiling panel. The arm lifts the panel to ceiling height, the participant walks in to work on it, and the arm then lowers it. Two separate systems track the participant.

Xsens MVN is a motion-capture suit. Seventeen small motion sensors, each with an accelerometer, gyroscope and magnetometer, are strapped to the body. Software combines their signals with a body model to give the position and orientation of 23 body segments, such as the pelvis, head, arms and legs, at up to 240 Hz [18], [19]. Because position is calculated by adding up small movements over time, it slowly drifts, and its starting point is set by a calibration pose rather than by the room. Here, Xsens provides the body-segment positions used for the body-distance input in the later version of the system.

OptiTrack is a camera-based motion-capture system. Infrared cameras around the cell (on the overhead frame in Fig. 3a) detect small reflective markers, and the Motive software locates each marker to about a millimetre in a fixed room frame [20]. A group of markers on the participant’s cap is tracked as one object. OptiTrack therefore gives the head’s position relative to the robot without drift, but says nothing about the rest of the body. Here, the OptiTrack head position is used to calculate the distance and closing speed that drive the phase model and the reactive limit.

The two systems have different jobs, and their estimates are not merged: OptiTrack gives the head’s position relative

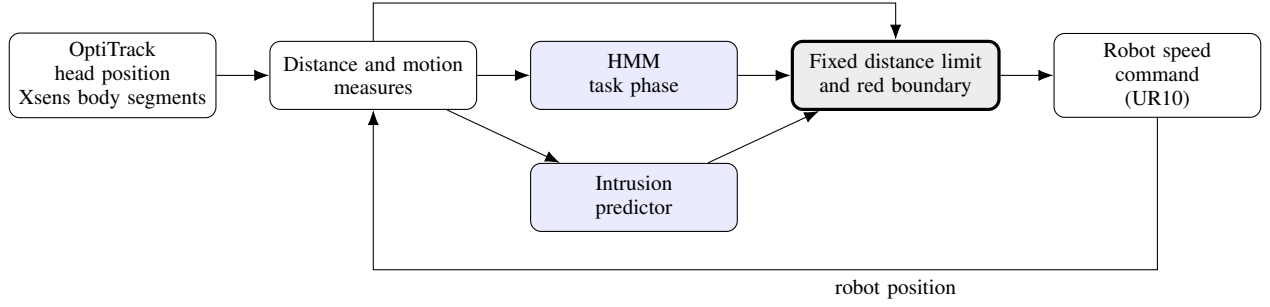


Fig. 2. How the prototype works. Motion capture and the robot’s position give the distance and motion measures. The learned parts (blue) can only slow the robot further; they cannot raise its speed above the fixed distance limit (grey). Data collection for the experiment is shown separately in Fig. 4.



Fig. 3. Lab equipment. (a) The cell: the UR10 arm holding the panel stand-in at ceiling height, OptiTrack cameras on the overhead frame, and the operator desk. (b) A participant wearing the Xsens suit and OptiTrack cap, seen from behind while walking toward the robot. (c) The working phase: the participant works under the raised panel. Faces are blurred.

to the robot, and Xsens gives body posture. Both feed the same processing step in the same format, so the controller does not depend on which system produced a sample. The robot controller reports the tool centre point (TCP), which is the position of the gripper face, and the speed of each of the six joints. The control loop runs at about 60 Hz, and the actual time between samples is logged.

C. Distance Measures and System Versions

The robot is represented by a vertical column, a simple stand-in for its shape. For the tracked head position $\mathbf{p}$ and the nearest point $\mathbf{q}$ on this column, the controller calculates

$$d = \|\mathbf{q} - \mathbf{p}\|_2, \quad v_{\text{proj}} = \mathbf{v}^T \mathbf{u}, \quad (1)$$

where d is the head-to-robot distance in metres, $\mathbf{v}$ is the head velocity estimated from recent samples, $\mathbf{u}$ is the unit vector pointing from the head toward the robot, and v_{proj} is the closing speed in m/s. A positive v_{proj} means the person is

moving toward the robot. The heading alignment h_t , used below, measures how directly the person’s movement points at the robot; it is not the direction the body faces. Table II lists all symbols.

Data were recorded with two versions of the system. (In this paper a code such as P17 is a study ID, and T05 is a trial within it; see Section V-B.) The 52 earlier records (P13–P22) used capture confirmed manually by the operator and the column stand-in for the robot. The 45 later records (P31–P38) streamed data automatically, used the robot’s live position, and used anchored body-segment positions for the body-distance input. The phase model used head-based measures in both versions. The two versions are reported separately in the measurement checks and are not treated as an experimental condition.

TABLE I
RELATED METHODS AND HOW THEY RELATE TO THIS STUDY.

Study	Method and test setting	What it means for this study
Byner et al. [8]	Speed limit set continuously from distance and direction of motion; tested in a physical machine-tending cycle.	Reactive speed adjustment is the baseline; phase recognition must add value on top of it.
Karagiannis et al. [9]	Safety zones switch with robot position; overhead safety camera and safety PLC in a car-industry cell.	A software speed limit is not the same as a system built from safety-rated parts.
X. Lin et al. [10]	Protective distance updated from worker, robot and material shapes; construction conflict scenarios.	Panel and body shapes need checking beyond whether a point-based distance is calculated consistently.
X. Lin et al. [14]	Learned work-step model and local safety control; VR wall-panel task with simulated sensors.	Task-aware safety control already exists; our focus is real robot data and participant experience.
This study	Real-time HMM task phase and motion prediction below a fixed speed limit; physical ceiling-panel prototype.	Robot data and questionnaires from a construction-style task with a person in the loop.

TABLE II
SYMBOLS USED IN THE CONTROLLER EQUATIONS.

Symbol	Meaning
d	Distance from tracked point to robot (m)
$v_{\text{proj}}, a_{\text{proj}}$	Speed (m/s) and acceleration (m/s ²) toward the robot
h_t	How directly the movement points at the robot
z_t	Task phase at time t (approaching, working, retreating)
$w_{ik}, \mu_{ik}, \sigma_{ik}^2$	Weight, mean and variance of component k in phase i
r_{red}	Radius of the inner (red) boundary (m)
τ	Predicted time to reach the red boundary (s)
S	Protective separation distance (m)
T	Assumed time to react and stop (s)
C	Margin for body parts not tracked (m)
S_a	Margin for sensor error (m)
$u_{\text{env}}, u_{\text{context}}, u_{\text{cmd}}$	Speed fractions (0–1) from the distance limit, from the learned parts, and sent to the robot

D. Task Phase and Intrusion Prediction

The model recognises three task phases: approaching, working and retreating. At each time step it uses four measures, $\mathbf{x}_t = [d_t, v_{\text{proj},t}, \|\mathbf{v}_t\|, h_t]^T$: distance, closing speed, overall speed and heading alignment. For each phase, the typical values of these measures are described by a mix of three Gaussian (bell-shaped) distributions:

$$p(\mathbf{x}_t | z_t = i) = \sum_{k=1}^3 w_{ik} \mathcal{N}(\mathbf{x}_t; \mu_{ik}, \text{diag}(\sigma_{ik}^2)), \quad (2)$$

where z_t is the phase at time t , w_{ik} is the weight of component k in phase i (the three weights add to one), and μ_{ik} and σ_{ik}^2 are that component's mean and variance for each measure. In real time the model can use only present and past data (the causal filter). For offline checking, the Viterbi method can also use later data, so results for the two are reported separately. Because the labelled phases are long and the model expects phases to persist, it changes phase slowly and may react late to short movements.

The intrusion predictor works independently of the HMM. It solves

$$d - r_{\text{red}} - v_{\text{proj}}\tau - \frac{1}{2}a_{\text{proj}}\tau^2 = 0 \quad (3)$$

for τ , the time the person would take to reach the inner (red) boundary of radius r_{red} from the current distance d if they kept their current closing speed v_{proj} and acceleration a_{proj} . If τ is between zero and 0.5 s, the controller flags an intrusion. The closing-speed threshold is 0.6 m/s, and the acceleration used is capped at ± 4 m/s². The predictor only detects movement toward the boundary; it does not guess the person's intention.

E. Controllers and Speed Command

Three controllers were compared. The *fixed* controller uses set distance bands: full speed, reduced speed and stop. The *reactive* controller also takes closing speed into account, using a simplified separation limit. The *predictive* controller adds phase-based caution and the intrusion predictor on top of the reactive limit. Its protective separation distance is

$$S = \max(0, v_{\text{proj}})T + C + S_a, \quad (4)$$

where S is the distance the controller tries to keep; T is the assumed time for the system to react and the robot to stop, so $v_{\text{proj}}T$ is how much closer the person gets in that time; C is a margin for body parts that the tracked point does not cover; and S_a is a margin for sensor error. This is the ISO/TS 15066 separation formula without the term for the robot's own movement [2]. We used $T = 0.4$ s, $C = 0.2$ m and $S_a = 0.1$ m. Beyond S , the allowed speed rises gradually over 0.3 m. The inner (red) radius is 0.94 m, and the yellow radius is 1.6 times that. These values were chosen for the lab cell; they are not measured stopping times or sensor errors of an installed system.

The speed sent to the robot is

$$u_{\text{cmd}} = \min(u_{\text{env}}, u_{\text{context}}), \quad 0 \leq u_{\text{cmd}} \leq 1, \quad (5)$$

where u_{env} is the speed fraction allowed by the reactive limit, u_{context} is the fraction suggested by the phase model and intrusion predictor, and u_{cmd} is the fraction sent to the robot. Taking the smaller of the two means the learned parts can only lower the speed, never raise it. Inside the red radius the robot is always asked to stop, and if tracking is lost the controller falls back to its most cautious setting. The command is sent through the robot's speed override and pause function. This is a research controller: its stop is not a safety-rated protective stop, and (5) limits the software request rather than certifying the installed system (Section VII-E).

Two details of the controller matter for the results. First, the HMM's phase caution can only cap the speed at the reduced level, so it never stops the robot on its own. Second, the intrusion predictor issues a stop once the risk has stayed above its threshold for two consecutive samples (about 33 ms at 60 Hz) and releases it as soon as the risk falls back below the same threshold; there is no separate, lower release threshold.

IV. MODEL DEVELOPMENT AND VALIDATION

We trained the phase model before the participant study, using 32 labelled trials in which the three team members acted as operators (study IDs P03, eight trials; P04, 12; P05, 12). These contain 90,487 frames: 17,267 approaching, 53,609 working and 19,611 retreating. We validated it by leaving one operator out at a time: the model was trained on two operators and tested on the third.

Two measures are reported. Accuracy is the share of frames whose predicted phase matches the label. Because most frames are "working", a model that favours that phase scores well on accuracy. Balanced accuracy averages the recall of the three phases, so each phase counts equally. Offline accuracy is 0.821 and real-time (causal) accuracy is 0.794 (Table III). Balanced accuracy is lower, especially for the causal filter (0.725), and recall is highest for the working phase. The small number of operators, the correlation between frames in the same trial, and the reuse of the same folds during development limit how far these values generalise.

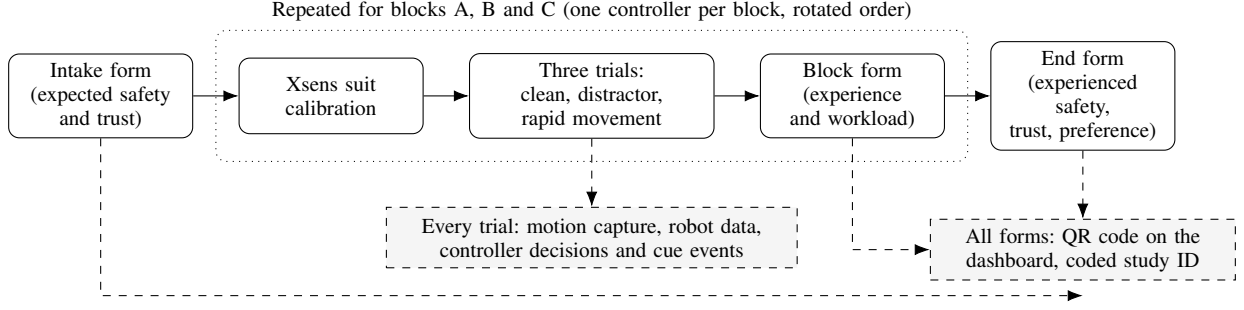


Fig. 4. One participant session. Solid arrows show the order of steps; dashed arrows show the data collected. The three controllers were assigned to blocks A, B and C in one of six rotated orders.

TABLE III
LEAVE-ONE-OPERATOR-OUT PHASE-RECOGNITION VALIDATION ON THE DEVELOPMENT SET. VITERBI DECODING USES THE FULL SEQUENCE; THE CAUSAL FILTER USES PRESENT AND PAST OBSERVATIONS.

Measure	Value
Development participants	3
Accepted trials	32
Labelled frames	90487
Offline Viterbi accuracy	0.821
Offline balanced accuracy	0.757
Causal-filter accuracy	0.794
Causal-filter balanced accuracy	0.725
Approaching recall	0.693
Working recall	0.923
Retreating recall	0.654
Offline fold P03 accuracy	0.812
Offline fold P04 accuracy	0.817
Offline fold P05 accuracy	0.832

V. EXPERIMENTAL METHOD

A. Task and Experimental Design

The protocol has three blocks, A, B and C. The fixed, reactive and predictive controllers were assigned to the blocks in one of six rotated orders. Each block contains one clean trial, one distractor trial and one rapid-movement trial (Fig. 4). Each trial follows the panel sequence: walk in and load the panel, step back before the lift, work under the raised panel, and step back before it is lowered and unloaded. The operator gave the scripted cue during the lift, so a cue window marks when the instruction was given, not when the movement started.

B. Participants and Analysed Records

The study ran under Monash University ethics approval (project 51642). The team recorded 12 people taking part, but because some sessions were restarted under a new P-number, that count cannot be rebuilt from the exported records alone. Study IDs therefore label recorded sessions, not confirmed individuals.

Because not every record could be linked to one person, the analysis is descriptive. The main data set is the 11 study IDs that have both intake and end forms and a known controller order. These IDs had 99 planned trial slots, for which 115 attempts were found: 108 completed and seven aborted. For each slot we used the latest completed trial whose files were

readable and whose sample count matched its record. This gave 97 trials; two slots (P13 block A trial 1 and P22 block A trial 3) had no completed trial. Earlier repeats of a slot are kept in the audit record but not analysed. The same 11 IDs gave 31 block questionnaires: nine IDs completed all three, and two are missing the last block. All 11 have matched intake and end forms. Robot data and questionnaires were analysed separately, and all values come directly from recorded samples and original form responses; no missing values were filled in.

C. Robot Data Processing

The selection above gave 97 trials and 494,075 samples. File fingerprints, row counts and time totals were rechecked independently after the analysis. For time totals, each sample's value is held until the next sample, but only when the gap to the next sample is between 0 and 0.25 s. Longer gaps are reported separately and left out of both the total and the time base. A robot-data sample counts as fresh when the packet is available, all six joint speeds are valid and the data are less than 0.25 s old. The robot counts as moving when its fastest joint exceeds 0.005 rad/s; counts at 0.001 and 0.01 rad/s are kept as a check on this threshold.

To check the head distance, we recalculated it from the same sample's head position and the robot's live tool position. This checks internal consistency, not physical calibration. For each cue window we report the smallest distance and the largest closing speed, which need not occur at the same moment. Stop time includes intentional holds in the task, so it is not a count of false stops.

The recorded stop counts were checked for a known system issue: the speed command sometimes switched rapidly between stop and run. To separate this switching from real stops, every stop request (a change of the command from above zero to zero) was classed by duration, as *brief* (under 0.5 s) or *sustained* (0.5 s or longer), and by the controller rule logged with it: sensing-dropout hold, intrusion predictor, red boundary or envelope minimum. The same rules were applied to all three controllers, and the raw counts reproduce the earlier totals for every trial. As a check, stops were also joined when separated by less than 0.25, 0.5 or 1.0 s and counted only if they lasted at least 0.5 or 1.0 s. No video was used.

D. Questionnaire Inclusion and Scoring

Pilot, link-test and qualification submissions were excluded. This left 18 intake forms, 31 block forms and 11 end forms, with 11, 11 and 9 block forms for A, B and C and no duplicates.

The block questionnaire uses five-point ratings. Three items run between two opposite words: anxious to relaxed, agitated to calm, and quiescent to surprised. Four items ask how much the participant agrees that they were confident the robot would stop, had to watch it constantly, felt it stopped more than needed, and found its speed changes made sense. Three items rate mental demand, physical demand and frustration. The workload items are adapted from NASA-TLX [16] and the confidence and monitoring items from Jian et al. [17]. Neither set is the full validated questionnaire, so each item is analysed on its own and no combined score is formed.

Block results are given as response counts and median [first, third quartile]. As a check, we repeat the summary using only the nine IDs with all three block forms, and count how many ratings went up, stayed the same or went down from A to C. These changes describe the order of the blocks; a higher rating is not always better. Results are grouped by block label, not by controller, because the assignment records alone do not prove which controller each questionnaire refers to. The intake and end forms are matched by study ID and ask about expected and then experienced safety and trust, from 1 (not at all) to 5 (completely). Because neither the number of independent people nor full controller exposure can be confirmed, we report no significance tests, confidence intervals or effect sizes, and we do not assume missing data are random.

VI. RESULTS

A. Robot Data

Each of the 11 study codes has eight or nine selected trials. Table IV summarises them by controller; these totals describe the records and are not estimates of a controller effect. The 97 trials contain 494,075 samples and 186.17 minutes of capture. 398 gaps are longer than 250 ms, totalling 281.65 s (2.52%); 10,888.35 s remain for time totals. 3244 samples are flagged as stale tracking, and 395 put the controller into its most cautious state. Controller decisions exist in 494,023 samples, and every one is recorded as sent to the robot. This confirms that commands were recorded, not how the robot physically stopped.

The trials are close to the planned 11 per trial type per controller; the two missing slots leave 10 clean fixed and 10 clean reactive trials. Median stop time is 29.7%, 33.6% and 36.7% for fixed, reactive and predictive records, and the raw numbers of stop requests are 659, 888 and 1417. Because controller order, system version and person identity are not jointly controlled, these differences are descriptive. Stop time also includes intentional task holds.

The safety indicators were similar across controllers (Table IV). Whenever the measured distance was inside the 0.94-m red boundary, all three controllers requested a stop 100% of the time. The median closest distance per trial was 0.08, 0.06 and 0.08 m, and people spent 28.5%, 31.3%

TABLE IV
RECORDED BEHAVIOUR BY CONTROLLER (97 TRIALS). BRIEF STOP REQUESTS LAST LESS THAN 0.5 s.

	Fixed	Reactive	Predictive
Trials	32	32	33
Clean / distractor / rapid	10/11/11	10/11/11	11/11/11
Median trial duration (s)	112.5	113.6	109.0
Median closest distance (m)	0.08	0.06	0.08
Time inside red boundary (%)	28.5	31.3	32.0
Stop requested inside red (%)	100	100	100
Median stop time (%)	29.7	33.6	36.7
All stop requests	659	888	1417
Brief: sensing dropout hold	450	414	326
Brief: intrusion predictor	0	0	805
Brief: other rules	43	293	69
Sustained: red boundary	164	156	101
Sustained: intrusion predictor	0	0	104
Sustained: other rules	2	25	12
Sustained stops per minute	2.7	3.1	3.6

Totals are descriptive, not controller-effect estimates; order, participant identity and system version are not jointly controlled. Stop time includes intentional task holds. P13 A1 and P22 A3 have no completed capture.

and 32.0% of observed time inside the red boundary, mostly during the loading, working and unloading approaches the task requires. Median trial duration was 112.5, 113.6 and 109.0 s, so predictive trials were not slower overall, although the robot spent more of that time stopped.

Most stop requests were brief: 493 of 659 for the fixed controller, 707 of 888 for the reactive controller and 1200 of 1417 for the predictive controller lasted less than 0.5 s (Table IV). Two causes account for most of them. The sensing-dropout hold, which pauses the arm when fresh Xsens body data are briefly missing and is shared by all three controllers, produced 450, 414 and 326 brief requests. The intrusion predictor, used only by the predictive controller, produced 805 brief and 104 sustained requests. Sustained stops were much closer between controllers: 166, 181 and 217, or 2.7, 3.1 and 3.6 per minute of observed time, and most were red-boundary stops during the approaches the task requires. Joining stops less than 0.5 s apart and counting only those of at least 1 s gave the same order (140, 153 and 187).

In the 33 later trials, the log also records what all three controllers would have requested at each sample from the same inputs. On these identical inputs the predictive controller started a stop 1.89 times as often as the reactive controller (median per trial; interquartile range 1.61–2.43), so the switching comes from the controller itself rather than from differences in participants or order. These same-input values are open-loop and do not show how the person or robot would then have moved. The switching also grew over the study: predictive trials logged a median of about 49 Hz on 9–10 September and 38 Hz on 16–17 September, while brief intrusion-predictor stops rose from about 6 to 20 per minute (rank correlation -0.72 , $n = 33$). Date, data rate and system version changed together, so their effects cannot be separated. Fig. 6 shows the switching in one trial.

Across the scheduled cue windows, fresh robot data show joint motion for 1712.80 s of observed time, confirming that the robot moved during the labelled event windows. Of 94,031

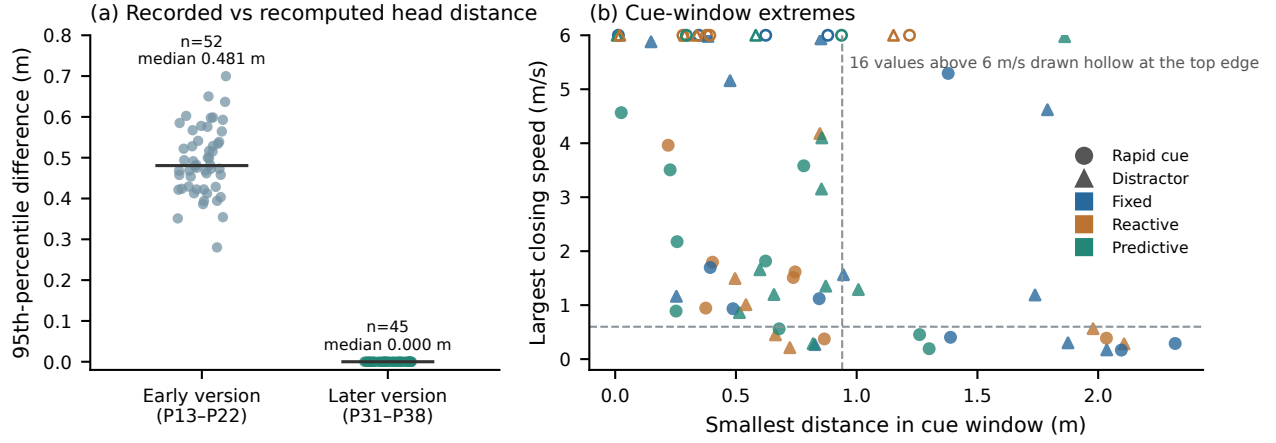


Fig. 5. Measurement checks, one mark per trial. (a) 95th-percentile difference between the recorded head distance and a recalculation from the same sample, by system version. (b) For each distractor or rapid cue window, the smallest distance (horizontal) against the largest closing speed (vertical). Dashed lines show the 0.94-m inner boundary and the 0.6-m/s closing threshold. Values above 6 m/s are physically implausible input spikes and are drawn hollow at the top edge. The two extremes need not occur at the same moment.

fresh samples in these windows, 79,356, 77,862 and 77,492 show motion at 0.001, 0.005 and 0.01 rad/s, so the result does not depend on the chosen threshold.

B. Distance Inputs and Cued Movements

In the early records, the median difference between the recorded head distance and the same-sample recalculation (95th percentile per trial) is 0.481 m (Fig. 5a). In the later records it is 0.000 m at the reported precision. This shows that the later logging is arithmetically consistent, not that it is physically calibrated or safety-rated.

The wide spread and overlap of rapid and distractor cue windows (Fig. 5b) show that the scripted cue name does not decide the movement that was actually measured. These values are controller inputs, not checked physical clearances, so the measured distance and closing speed, not the cue name, must guide interpretation. Figure 6 shows a typical later record. Because the cue marks the instruction rather than the start of the movement, the delay between cue and robot response is not reported as a protective response time.

C. Reported Experience

Block responses lean toward relaxed and calm, confidence that the robot would stop, and low demand and frustration (Table V). Median confidence that the robot would stop is five in every block, while median agreement that it stopped more than needed is three in A and two in B and C. This is a perception, not an objective count of false stops, and responses are clustered within study IDs.

Using only the nine complete response sets keeps 29 of the 30 block medians unchanged; the exception is constant monitoring in B, which changes from two to one. The paired A–C counts show variation that the medians hide (Table VI). Five relaxed ratings increase and four stay the same; none decrease. Ratings of unnecessary stopping increase twice, decrease twice and stay the same five times, despite the lower median in C. Surprise decreases for three codes, increases for

two and is unchanged for four. Lower block medians therefore do not mean that everyone improved.

Across the 11 matched records, the mean safety rating was 3.82/5 before and 4.45/5 after the session: eight ratings increased, one was unchanged and two decreased. Mean trust was 4.00 before and 4.18 after: five increased, three were unchanged and three decreased (Table VII). The intake form asks about expected and the end form about experienced safety and trust, and individual linkage is incomplete, which limits how these changes can be read.

In the end forms, block C was chosen as safest seven times, compared with two each for A and B. For a full work shift, six chose C, three B and two A. Choices for least safe were spread across all blocks (A three, B four, C four). Because the link between block label and controller is not fully established, C is not read as the predictive controller. Four participants noticed changes between blocks and seven did not; their comments refer to stopping or slowing sensitivity and lift speed. The median rating for how much the suit restricted movement was two (interquartile range 1.5–3), and no end form reported risk or discomfort.

VII. DISCUSSION

A. Context-Aware Control in the Panel Task

The prototype combines the task phase and motion prediction while keeping a fixed software limit on the commanded speed. This separates two questions: whether a movement belongs to a normal task phase, and whether the measured movement is approaching the boundary. A wrong HMM phase cannot allow a higher speed than the distance limit, although errors in geometry or timing still affect that limit.

The validation results show that phase recognition is strongest in the longer working phase. Because balanced accuracy in real time is lower than overall accuracy, phase accuracy alone does not show practical benefit. The predictive controller's higher raw stop count is mainly a switching effect, not more or longer holds. Of its 1417 stop requests, 805 were

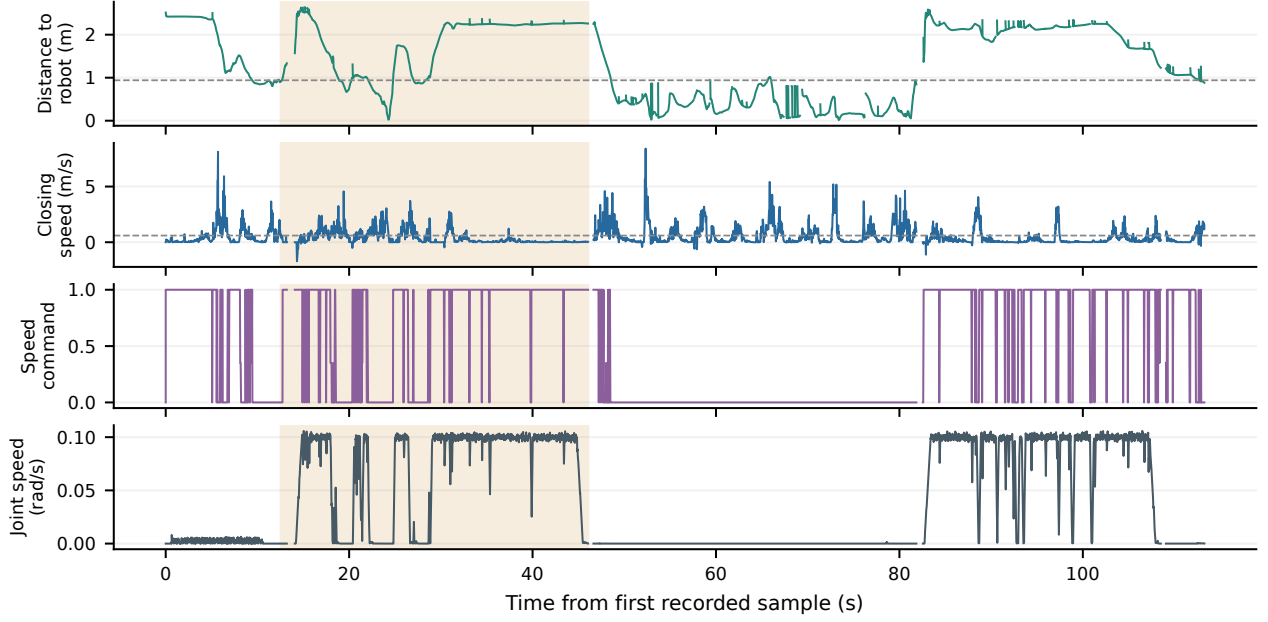


Fig. 6. A typical later trial (P31 T10, predictive controller): distance to the robot, closing speed, speed command and fastest joint speed. Shading marks the cue window; dashed lines are the configured thresholds; gaps over 250 ms break the lines. Note the rapid switching of the speed command.

TABLE V
BLOCK QUESTIONNAIRE ITEMS: MEDIAN [FIRST, THIRD QUARTILE]. A, B AND C ARE REPORTED BLOCK LABELS, NOT POOLED CONTROLLER CONDITIONS. ALL RATINGS USE 1–5.

Item	A ($n = 11$)	B ($n = 11$)	C ($n = 9$)
Anxious to relaxed	4 [3.5, 4.5]	5 [4.0, 5.0]	5 [4.0, 5.0]
Agitated to calm	4 [4.0, 5.0]	5 [4.0, 5.0]	5 [4.0, 5.0]
Quiescent to surprised	3 [1.0, 3.0]	3 [1.0, 4.0]	1 [1.0, 3.0]
Confident robot would stop	5 [4.0, 5.0]	5 [4.0, 5.0]	5 [4.0, 5.0]
Had to watch constantly	2 [1.0, 3.0]	2 [1.0, 3.0]	1 [1.0, 3.0]
Stopped more than needed	3 [2.0, 4.0]	2 [1.0, 3.5]	2 [2.0, 3.0]
Speed changes made sense	5 [3.5, 5.0]	5 [4.0, 5.0]	4 [4.0, 5.0]
Mental demand	1 [1.0, 3.0]	1 [1.0, 2.0]	1 [1.0, 2.0]
Physical demand	1 [1.0, 3.0]	1 [1.0, 2.5]	1 [1.0, 3.0]
Frustration	1 [1.0, 3.0]	1 [1.0, 2.0]	1 [1.0, 1.0]

Higher means the right-hand semantic anchor for the first three items, stronger agreement for the next four, and greater demand/frustration for the final three. Responses are clustered within 11 study IDs; individual-level linkage is incomplete. No trust, safety or workload composite is used.

TABLE VI
COMPLETE RESPONSE SETS: BLOCK MEDIAN AND WITHIN-CODE CHANGES ($n = 9$ CODES).

Item	A/B/C	A to C: up/same/down
Anxious to relaxed	4/5/5	5/4/0
Agitated to calm	4/5/5	3/6/0
Quiescent to surprised	3/3/1	2/4/3
Confident robot would stop	5/5/5	3/4/2
Had to watch constantly	2/1/1	1/5/3
Stopped more than needed	3/2/2	2/5/2
Speed changes made sense	5/5/4	1/5/3
Mental demand	1/1/1	2/6/1
Physical demand	1/1/1	1/6/2
Frustration	1/1/1	1/5/3

Counts refer to the raw rating direction, not uniformly to improvement. A–C changes describe block sequence, not a controller effect. The same nine codes supply every row.

TABLE VII
PAIRED GLOBAL RATINGS FOR 11 CODES WITH INTAKE AND END RESPONSES. FIVE-POINT ITEMS; DESCRIPTIVE CODE-LEVEL CHANGES ONLY.

Item	Before	After	Δ	Up/same/down
Felt safe	3.82	4.45	0.64	8/1/2
Trust in distance	4.00	4.18	0.18	5/3/3

Before/after wording changes from expected to experienced safety/trust. Means summarize coded responses; neither causation nor an interval-scale measurement model is assumed.

brief requests from the intrusion predictor; the HMM caused none, because its caution can only reduce speed. The predictor

stops the robot after two samples above its threshold and releases as soon as the risk drops, so when a noisy closing-speed estimate hovers near the threshold, the command flips between stop and run. The lower and less even data rate on the later days makes the closing speed noisier and is consistent with the rise in switching. A release delay, or a release threshold set below the trigger threshold (hysteresis),

should remove most of this switching without touching the fixed distance limit. A simple minimum stop time is not a free fix: replaying the recorded predictive commands with a 1-s minimum stop cut stop requests from 1417 to 420 but raised stopped time from 1351 to 2064 s. The release rule therefore needs testing on live runs.

The reactive controller remains the key reference for any future controlled comparison. Equation (5) also limits the expected benefit: for the same inputs, settings and zone, the predictive controller can never request a higher speed than the reactive one, because prediction and phase caution can only keep or lower the request. Any benefit must therefore come from the timing of intervention or from changes in how the person and robot then coordinate, not from a more permissive speed limit. Such a benefit cannot be shown from phase accuracy or by replaying different controllers on the same recorded movement.

Against H1, the results split into two parts. On safety, we found no sign that the predictive controller was less safe than fixed-zone control: a stop was requested every time a person was inside the red boundary, closest approaches and time inside the boundary were similar, the learned parts can only lower the speed (Equation (5)), and confidence that the robot would stop was high in every block. On efficiency, trials took about the same time (median 109.0 against 112.5 s), but the predictive controller kept the robot stopped for more of the trial (36.7% against 29.7%) and had more sustained stops (3.6 against 2.7 per minute), mostly because of the intrusion predictor. H1 is therefore partly supported: the safety part is consistent with the data, but comparable robot availability was not shown. Because participant, order and system version were not controlled, and equal safety cannot be formally established (Section VII-C), H1 remains inconclusive rather than confirmed or rejected. Fixing the predictor's switching is the clearest route to closing the efficiency gap.

B. Measured Behaviour and Reported Experience

The robot data show how the distance input, speed command and joint motion change through the panel sequence. The difference between the two system versions strongly affects how these signals can be read. Likewise, the gap between scheduled rapid movements and the measured closing speeds shows that event labels cannot replace measured exposure. Both findings support the need to validate the whole chain from sensing to stopping when SSM is implemented [3].

The questionnaires generally show confidence that the robot would stop, low mental and physical demand, and little frustration. Experienced safety was higher than expected safety in the matched records, while the change in trust was smaller. This suggests a positive experience of the lab task. However, novelty, supervision, repeated exposure and expectations could all contribute, so the change cannot be credited to the predictive controller.

The complete-response check makes the overall pattern less dependent on the two missing final-block forms, but it does not remove bias from missing data or solve the linkage problem. Very high ratings for stopping confidence and very

low ratings for demand and frustration also leave little room to tell the controllers apart. The paired results for stopping and surprise show why medians should be read alongside changes within each code. Growing familiarity may explain why people relaxed over the blocks, but this was not tested.

The preference for block C does not rank the controllers, because block labels rotate across controllers and full exposure linkage is not available. Likewise, agreeing that the robot stopped more than needed tells us about perceived interruption, not an objective false-stop rate. Reported and measured results describe different sides of collaboration and should be read together.

C. Safety and Efficiency

Research on adaptive SSM motivates measuring robot availability and cycle performance [8], [9]. This study builds what is needed to measure those outcomes but does not show a safety or efficiency advantage. Stop time combines intentional holds and responses to people nearby, and capture time includes operator actions, so it is not the same as task time. These measures therefore do not establish false stops, task efficiency or collaboration fluency [15]. Formally showing that the predictive controller is as safe as fixed-zone control would also need a justified equivalence margin, validated protective geometry and measured end-to-end stopping response, none of which this analysis provides. The safety indicators are therefore consistent with H1 but do not prove equal safety.

D. Coverage of the Planned Measures

The study design planned four system measures (human-robot distance, number of slowdowns and stops, number of yellow and red zone entries, and task completion time) and four experience measures (safety, comfort, predictability and trust) to compare fixed and adaptive safety. Distance and stops are reported per controller (Table IV, Figs. 5 and 6); red-boundary stops and part of the stop time come from the loading and unloading approaches the task requires, so they are not counts of unnecessary interruption. Trial duration includes operator actions, so it stands in for task completion time. Safety and trust are reported as intake and end ratings, comfort through the relaxed and calm items, and predictability through the speed-change item.

E. Limitations

The sample is small and was recruited for convenience. Because sessions were restarted, not every study ID can be linked to one person, so the 11 study codes cannot be treated as 11 independent people. Two trial slots lack a completed capture and two questionnaire sets lack block C, and missing data may relate to technical problems or task execution. The system version changed during collection, and the Xsens data rate fell on later days. Head points and body-segment positions do not cover the full body, tool and panel, and agreement with a calculation does not prove physical calibration. The controller is a research prototype: the separation limit is simplified, the command uses the ordinary robot speed override

rather than a safety-rated stop, and the end-to-end response was not measured, so the results do not support a certified safety claim. Cued movements, operator pacing, visible robot motion, learning and fatigue may affect both behaviour and ratings. The phase model was built from three operators and may not generalise to construction workers.

F. Future Work

Future controlled tests should keep one system version, check its sensor error against the full moving volume, and add a release delay or hysteresis to the intrusion predictor. Synchronised measurement of body movement, controller output and robot braking would give response times that do not depend on operator cues, and a study ID that follows each person across restarted sessions would allow verified within-person comparison of the controllers.

VIII. CONCLUSION

This paper presented a speed and separation monitoring controller that uses the task phase in human–robot ceiling-panel installation, and tested it on a physical prototype with human participants. The controller combines measured closing speed, recognition of three task phases and prediction of fast approaches, all below a fixed speed limit, so the learned parts can only make the robot more cautious. In real time, the phase model was correct 79.4% of the time on operators it had not been trained on. Across 97 robot-data trials and 31 block questionnaires, participants were confident that the robot would stop, reported low workload, and felt safer by the end of the session. Using only complete response sets left 29 of the 30 block medians unchanged, but the paired results showed that lower median ratings of unnecessary stopping did not hold for every participant. H1, that the predictive controller is as safe as fixed-zone control while keeping comparable efficiency, was partly supported: every controller requested a stop whenever a person was inside the red boundary, closest approaches were similar and trials took about the same time, but the robot spent more time stopped under predictive control, mostly because of brief switching in the intrusion predictor, which makes hysteresis the next design change. Formally showing equal safety and comparable availability will need a consistent distance measurement, synchronised measurement of the robot’s response, and an ID that follows each participant across restarted sessions.

DATA AVAILABILITY

Analysis scripts, derived tables and figure-generation code accompany the manuscript. Participant-level data require controlled access. All reported summaries are calculated from recorded observations without imputation or simulated values.

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AI ASSISTANCE DECLARATION

AI tools assisted with data-processing and analysis code, figure and table generation, and drafting and editing of the manuscript text.

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